

Input / Output / Process

• Output: print (

things you want to print

↳ A function

e.g. print("Hello World!")

Output Hello World!

String: A data type for text

↳ To create a string you must include

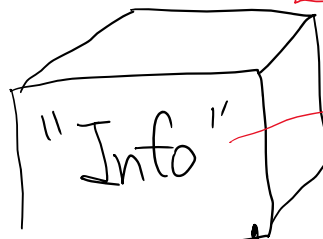
- Single Quotation Mark ('Hello')
- Two Quotation Marks ("Hello")

↳ e.g. print("I'm fine.")

* Variables *

↳ A way to store and reuse information/data

↳ Syntax: var_name = "Info"



store

* If you call variable without value, it will give error.

"IT I."



How to use: $\rightarrow \text{print}(\text{var_name}) \rightarrow \text{print}(\text{"Info"})$
 $\hookrightarrow$ When the program encounters a variable, it will replace the variable with the value inside the variable. $\hookrightarrow \text{Info}$

EX 1

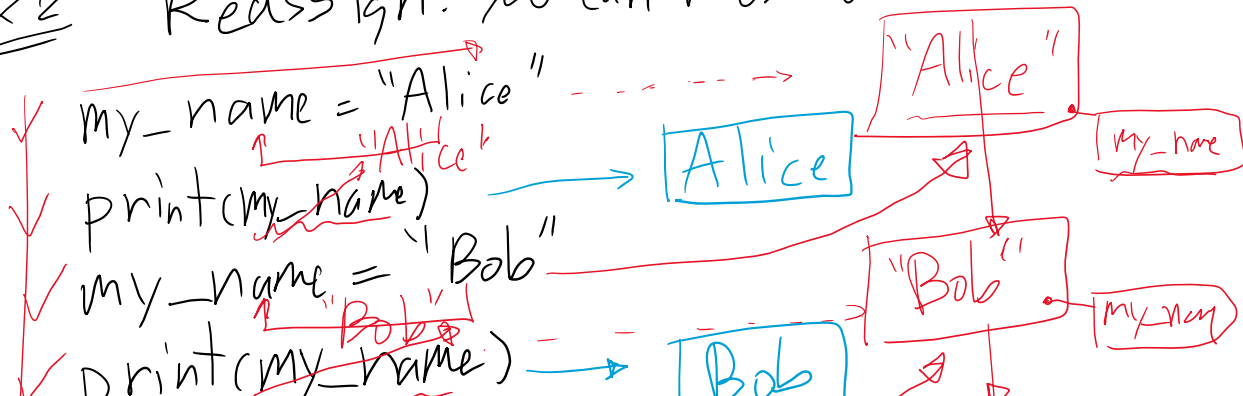
✓ $\text{my_name} = \text{"Nut"} \rightarrow \text{"Nut"}$ (my_name)

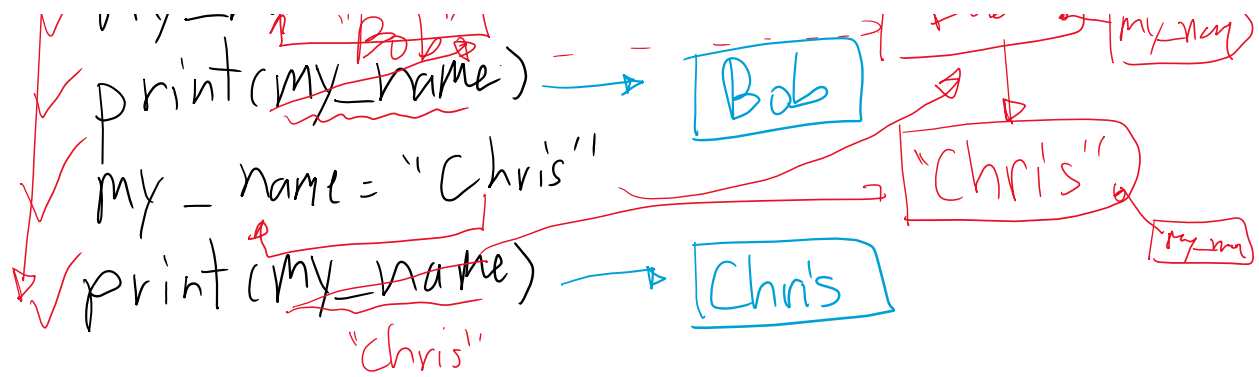
✓ $\text{print}(\text{my_name}) \rightarrow \text{Nut}$

Variable Naming Rules

- 1) Cannot contain spaces
 $\hookrightarrow$ e.g. my_name ✓ my name ✗
- 2) The first character must be letter or $_$.
 $\hookrightarrow$ e.g. 123char ✗
- 3) Cannot be python's keywords
- 4) Suggestion: Name your variable meaningful
 $\hookrightarrow$ e.g. not meaningful sdf , xd , lol , xYzX

EX 2 Reassign: you can reuse variables





Variable / data types

<u>String</u>	<u>Integer</u>	<u>Float</u>
text	numbers without decimal	real numbers
"Hello", "Hi"	123, -10, 0	0.12, -10.32, 0.1
e.g.	num = 1 print(num) → 1 num = 2 print(num) → 2	

Math Operators

- + plus
- - minus

→ Long Division

$$\begin{array}{r} 13 \\ 5 \overline{) 68} \end{array}$$

• * multiply

• / divide

• // Integer division

• % remainder / modulo

↳ e.g. $68 // 5 \rightarrow 13$

$68 \% 5 \rightarrow 3$

↳ $68 = 13 * 5 + 3$

↳ Ex $7 \% 2 = 1$

$7 // 2 = 3$

• ** exponential

↳ $a ** b \rightarrow a^b$

$\sqrt{a} \rightarrow a ** (1/2)$

$$\begin{array}{r} 5 \overline{) 68} \\ 5 \\ \hline 18 \\ 15 \\ \hline 3 \end{array}$$

$$\begin{array}{r} 2 \overline{) 7} \\ 6 \\ \hline 1 \end{array}$$

Math Operator Order

1) (), 2) **, 3) *, /, //, %, 4) +, -

Ex1: $1 + 2 * 3$

$1 + 6 \rightarrow 7$

Ex2: $2 + 1 * 3 + 10 // 4$

$2 + 3 + 2 \rightarrow 7$

EX3: $5 * 2 ** 3 + 1$

$\downarrow$
 $5 * 8 + 1$
 $40 + 1 \rightarrow \boxed{41}$

EX4: $(10 + 2) * (10 + 3) ** 2$

$12 * 13 ** 2$
 $\downarrow$
 $12 * 162 \rightarrow \dots$

Math Equation to Code

$$y = \frac{3x}{2} \rightarrow y = 3 * x / 2$$

$$z = 3bc + 4 \rightarrow z = 3 * b * c + 4$$

$$a = \frac{x + 2}{b + 1} \rightarrow a = (x + 2) / (b + 1)$$

print multiple things

↳ you can use commas to add more things to print one line.

↳ will add a space between things.

e.g.

$x = 1$

$y = 2$

$\text{print}(x, y) \rightarrow$

$\boxed{1 \ 2}$

$x = \text{"Hello"}$

$y = 2$

$\text{print}(x, y, x) \rightarrow$

$\boxed{\text{Hello} \ 2 \ \text{Hello}}$

e.g.

✓ $x = 0$

$\xrightarrow{x+1 \neq 0} \boxed{0} \xrightarrow{+1} x$

e.g. $\checkmark X = 0$
 $\checkmark X = X + 1$ $\rightarrow 0 + 1 = 1$
 $\checkmark X = X + 1$ $\rightarrow 1 + 1 = 2$
 $\checkmark X = X + 1$ $\rightarrow 2 + 1 = 3$

print(X , $X * 2$) $\rightarrow$ 3 6

$\downarrow$ $\downarrow$
 3 $3 * 2$

Input

$\hookrightarrow$ input (Optional text) to display to the user.
string

$\hookrightarrow$ ask the user to type something and when the user 'enters', it will return the value to the program.

$\hookrightarrow$ substitute the input() with that value
 * input will return a string

e.g. $X = \text{input}(\text{"Enter something: "})$
 $\uparrow$ the user enters 123.
 "1 2 3"

 `x = input("Enter something: ")`

... Enter something:

T.

Turn string to number

$\text{int}(\text{string})$: turn a string to integer

$\text{float}(\text{string})$: turn a string to float

↳ e.g. $\text{int}("123") \rightarrow 123$

$\text{float}("10.1") \rightarrow 10.1$

$x = \text{input}(\text{"Enter x: "})$ $\swarrow$ "10"

$\{ \quad x = \text{int}(x) \rightarrow \text{int}("10") \rightarrow 10$

$\text{print}(x+5) \rightarrow$ 15 $\swarrow$ "10"

$\downarrow$
 $10 + 5 = 15$

$x = \text{int}(\text{input}(\text{"Enter X: "}))$

$\underbrace{\hspace{10em}}_{\text{int}("10") \rightarrow 10}$

$\text{print}(x) \rightarrow$ 10